Symmetry Averaging and Hidden Information

A finite-support Shannon analysis of projection, recursion, and recovery

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Abstract

A constant observation need not imply a deterministic underlying state. We give a self-contained finite-support analysis of this distinction for symmetry averaging. A self-adjoint involution induces complementary orthogonal projections onto its symmetric and antisymmetric subspaces. For a random vector, the entropy hidden by the symmetric projection is exactly the conditional entropy of the residual given the observation. A two-dimensional example has a constant output while hiding an arbitrarily large finite number of bits. We prove that iteration of the projection stabilizes after one step, characterize the side information needed for exact recovery, and distinguish irreversible state replacement from passive observation of a state that continues to evolve. A finite-horizon deterministic example reveals one initially hidden bit without generating new randomness. The results are elementary consequences of linear algebra and Shannon entropy identities; no novelty claim is made for these underlying facts. The contribution is a unified exposition with explicit assumptions and a precise boundary between mathematical statements and interpretive language.

Keywords: Shannon entropy; orthogonal projection; symmetry averaging; hidden information; deterministic dynamics; exact simulation.

1 Purpose and scope

The motivating illustration uses two basis labels, “logic” and “nonsense”, and averages over their interchange. Such labels do not turn linear algebra into a logic admitting contradictions. Nor does a noninjective observation establish a physical source of randomness. This note replaces those interpretations by statements about maps, finite probability distributions, and reconstructibility.

Shannon entropy is used in its discrete sense [1]. Symmetry averaging is a standard projection construction; finite-group averaging also appears in contemporary applications [2]. All results needed here are proved explicitly. The term *recursion* refers solely to iteration of a stated map. The note does not establish a new coding theorem, a model of consciousness, or a physical implementation of the operators.

2 Conventions

Let $\mathcal{H}$ be a finite-dimensional complex Hilbert space. Every random variable in the note has finite support. A random vector in $\mathcal{H}$ is therefore a classical discrete random variable whose distinct values happen to be vectors. Its entropy is

$$H(X) = -\sum_{x \in \operatorname{supp} X} p_X(x) \log_2 p_X(x), \qquad 0 \log_2 0 := 0.$$

Entropy is measured in bits. No differential entropy or von Neumann entropy is used. Vectors are not assumed normalized, and equality of vector values is not equality of quantum rays.

3 The symmetry projection

Let $\Sigma : \mathcal{H} \to \mathcal{H}$ satisfy

$$\Sigma^* = \Sigma, \qquad \Sigma^2 = I, \qquad P = \frac{I+\Sigma}{2}, \quad Q = \frac{I-\Sigma}{2}. \tag{1}$$

Proposition 1 (Complementary projections)**.** *The operators P, Q are orthogonal projections, $P+Q = I$, and $PQ = QP = 0$. Moreover,*

$$\operatorname{Ran} P = \ker(\Sigma - I), \qquad \operatorname{Ran} Q = \ker(\Sigma + I).$$

Proof. Self-adjointness follows from that of Σ. Direct multiplication gives $P^2 = P$, $Q^2 = Q$, and $PQ = QP = (I - \Sigma^2)/4 = 0$. Also $\Sigma P = P$, while $\Sigma v = v$ implies $Pv = v$. These two observations identify $\operatorname{Ran} P$. Replacing the sign proves the assertion for Q. □

For the two-element group represented by $\{I, \Sigma\}$, P is the Reynolds average. More generally, a unitary representation U of a finite group G has average $P_G = |G|^{-1} \sum_{g \in G} U_g$. Inversion permutes G, giving $P_G^* = P_G$; counting the $|G|$ pairs with product k gives $P_G^2 = P_G$. Left multiplication permutes G, so its range is exactly the invariant subspace.

3.1 The two-dimensional motivating model

Choose an orthonormal basis e_0, e_1 of $\mathbb{C}^2$, optionally labelled "logic" and "nonsense", and set

$$\Sigma e_0 = e_1, \quad \Sigma e_1 = e_0, \qquad \psi_+ = \frac{e_0 + e_1}{\sqrt{2}}, \quad \psi_- = \frac{e_0 - e_1}{\sqrt{2}}.$$

Then

$$P\psi_+ = \psi_+, \quad P\psi_- = 0, \qquad P(ae_0 + be_1) = \frac{a+b}{\sqrt{2}}\psi_+. \tag{2}$$

Thus $\operatorname{Ran} P = \mathbb{C}\psi_+$ and $\ker P = \mathbb{C}\psi_-$. For $a + b \neq 0$, the output has ray $[\psi_+]$; its norm need not be one. For $a + b = 0$, the output is zero and has no projective ray. This exceptional case is essential.

4 Exact accounting of hidden entropy

For a finite-support random vector Z, define the observation and residual by

$$Y = PZ, \qquad R = QZ. \tag{3}$$

Theorem 2 (Projection–residual identity)**.** *The map $z \mapsto (Pz, Qz)$ is injective, with inverse $(y, r) \mapsto y + r$ on its image. Consequently,*

$$\boxed{H(Z) = H(Y, R) = H(Y) + H(R \mid Y), \qquad H(Z \mid Y) = H(R \mid Y).} \tag{4}$$

In particular, $H(Y) \leq H(Z)$, with equality exactly when Z is a deterministic function of Y almost surely.

Proof. Since $P + Q = I$, $Z = Y + R$. The pair (Y, R) is a deterministic function of Z and conversely, so their nonzero atom probabilities coincide up to relabelling. Thus their entropies agree. The chain rule gives $H(Y, R) = H(Y) + H(R \mid Y)$. Since Y is a function of Z, the same rule gives $H(Z) = H(Y) + H(Z \mid Y)$. Subtraction proves the conditional identity. A finite conditional entropy vanishes exactly when each positive-probability conditional distribution is concentrated at one value, which is equivalent to deterministic recovery. □

5 A constant observation with arbitrary hidden entropy

Let W be any finite-support complex-valued random variable and set

$$Z = \psi_+ + W\psi_-, \qquad Y = PZ = \psi_+, \qquad R = QZ = W\psi_-. \tag{5}$$

Corollary 3 (Constant readout)**.** *For the model* (5)*,*

$$H(Y) = 0, \qquad I(Z;Y) = 0, \qquad H(Z) = H(R) = H(Z \mid Y) = H(W).$$

Proof. The output is constant. Both maps $w \mapsto \psi_+ + w\psi_-$ and $w \mapsto w\psi_-$ are injective because $\psi_- \neq 0$. They preserve atom probabilities. The remaining conclusions follow from Theorem 2. □

For $W \in \{-1, 1\}$ with $\Pr(W = 1) = p$, the hidden entropy equals

$$h_2(p) = -p \log_2 p - (1-p) \log_2 (1-p).$$

At $p = 1/2$ it is one bit; at $p = 0$ or 1 it is zero. The projection alone does not select p. If W is uniform on m distinct values, the hidden entropy is $\log_2 m$ bits. Hence even a one-dimensional kernel can hide an arbitrarily large finite entropy when exact distinguishability and unrestricted finite support are allowed. A finite measurement resolution or an amplitude constraint requires a separate observation model.

6 Recursion: stabilization is not randomness generation

Proposition 4 (One-step stabilization)**.** *For $Z_0 = Z$ and $Z_{n+1} = PZ_n$, one has $Z_n = PZ$ for every $n \geq 1$. Therefore*

$$H(Z_n) = H(PZ) \quad (n \geq 1), \qquad H(Z_{n+1} \mid Z_n) = 0 \quad (n \geq 0).$$

Proof. Idempotence gives $P^n = P$ for $n \geq 1$. Each successive state is a deterministic function of the preceding one, so its conditional entropy is zero. □

If PZ replaces the entire stored state and the residual is discarded, the lost distinction cannot be recovered from repeated application of P. In contrast, an architecture retaining (PZ, QZ) preserves Z exactly. The algebraic identity $Z = PZ + QZ$ alone does not assert that a physical device actually stores QZ.

7 Side information and exact recovery

Theorem 5 (Conditional side-information requirement)**.** *Suppose a finite-valued variable M permits exact recovery $Z = d(Y, M)$ almost surely. Then*

$$H(M \mid Y) \geq H(Z \mid Y) = H(R \mid Y). \tag{6}$$

The choice $M = R$ attains equality.

Proof. Exact recovery implies $H(Z \mid Y, M) = 0$. Applying the conditional chain rule in two orders yields

$$H(M \mid Y) = H(Z, M \mid Y) = H(Z \mid Y) + H(M \mid Z, Y) \geq H(Z \mid Y).$$

For $M = R$, reconstruction is $d(y, r) = y + r$, and Theorem 2 gives equality. □

This is an entropy statement about random side information. It is not an assertion that a one-shot code has length exactly $H(R \mid Y)$, nor that this number measures energy consumption.

8 Deterministic dynamics can reveal hidden distinctions

Passive observation differs from replacing the state by its projection. Let $B : \mathcal{H} \to \mathcal{H}$ be linear, retain the full state, and observe

$$Y_t = PB^t Z, \qquad t = 0, 1, \ldots. \tag{7}$$

Proposition 6 (Criterion for a closed observed update)**.** *There exists a linear $T : \operatorname{Ran} P \to \operatorname{Ran} P$ satisfying $PB = TP$ if and only if*

$$PBQ = 0. \tag{8}$$

If this holds, $Y_{t+1} = TY_t$. If it fails, there exist two states with the same current observation and different next observations.

Proof. If $PB = TP$, then $PBQ = TPQ = 0$. Conversely, if $PBQ = 0$, then $PB = PBP$. Set $T = PB|_{\operatorname{Ran} P}$ to obtain $PB = TP$. This identity applied to $B^t Z$ gives the observed recursion. If $PBQ \neq 0$, choose v with $PBQv \neq 0$. The states z and $z + Qv$ have identical projections, but their next projections differ by $PBQv$. □

A nonzero coupling PBQ permits hidden distinctions to become visible, but does not by itself specify their probability law.

8.1 An explicit one-bit example

Let W be a fair random sign, $Z = \psi_+ + W\psi_-$, and let B swap the symmetric and antisymmetric vectors:

$$B\psi_+ = \psi_-, \qquad B\psi_- = \psi_+.$$

Then $B^2 = I$ and

$$Y_0 = \psi_+, \qquad Y_1 = W\psi_+, \qquad Y_{2k} = \psi_+, \quad Y_{2k+1} = W\psi_+.$$

Thus

$$\boxed{H(Y_1 \mid Y_0) = 1, \qquad H(Y_1 \mid Z) = 0.} \tag{9}$$

For every horizon $n \geq 1$, $H(Y_0, \ldots, Y_n) = 1$. The first nonconstant observation reveals the initial bit; subsequent observations do not introduce fresh independent bits. Here ψ_+ and $-\psi_+$ are distinct classical vector outputs. They would represent the same quantum ray, so this example must not be reinterpreted as a quantum phase measurement without changing the model.

8.2 An exact seed statement for a fixed transcript

For a fixed horizon n, write $\mathbf{Y}^{(n)} = (Y_0, \ldots, Y_n)$ and define

$$C_n = \inf\{H(S) : S \text{ has finite support and } d(S) \overset{d}{=} \mathbf{Y}^{(n)} \text{ for a deterministic } d\}.$$

Proposition 7 (Seed entropy for one transcript law)**.** $C_n = H(\mathbf{Y}^{(n)})$*. In the fair-sign example, $C_0 = 0$ and $C_n = 1$ for $n \geq 1$.*

Proof. A deterministic map cannot increase discrete entropy, so every admissible seed satisfies $H(S) \geq H(\mathbf{Y}^{(n)})$. Equality is attained by taking S with the transcript distribution and using the identity decoder. The example follows from the preceding calculation. □

This statement allows an arbitrarily distributed finite seed. It is not a bound on the number of fair coin flips needed for sampling. It concerns a single prescribed transcript law, not a common causal decoder required to work simultaneously across adaptive policies. The latter requires its own definitions and theorem.

9 Interpretation, limitations, and research directions

What is established. The symmetric component and the residual form a lossless representation of the input. Retaining only the symmetric component loses precisely $H(R \mid Y)$ bits of Shannon information. A constant readout can conceal arbitrary finite entropy, while repeated projection creates no further randomness. Retained hidden information can become observable when subsequent dynamics couple the kernel of the readout into its range.

What the basis labels do not establish. The names "logic" and "nonsense" are interchangeable labels for orthogonal vectors. Their sum is not the logical conjunction of a proposition and its negation. The symmetric vector is a fixed point of an explicitly chosen map, not a resolution of a foundational logical paradox. The informal phrase "paradox recursion" can describe the motivating artwork, but does not replace the operator definitions.

No inferred probability source. The distribution of W is an input assumption. Neither a kernel, a projection, nor a geometric singularity supplies that distribution. The construction demonstrates conditional uncertainty in a classical model; it does not derive fundamental randomness or a physical hidden-variable theory.

No thermodynamic identification. The entropy differences above quantify distinguishability under a probability law. They are not a work or heat budget. Any thermodynamic claim would require a physical implementation, a state preparation and reset protocol, and additional assumptions.

Concrete extensions. A useful next step is to specify a finite-resolution readout $q_\delta(PB^tZ)$ and compute the remaining conditional entropy for a stated source law. For multiple interventions, one should first fix the controlled history tree and the admissible common-seed decoder class. Neither problem is settled by merely iterating an idempotent operator.

10 Conclusion

$$\boxed{H(Z) = H(PZ) + H\big((I-P)Z \mid PZ\big).}$$

A constant output says that the output is constant. Whether the missing distinction is discarded, retained elsewhere, or later revealed is a property of the complete model. Symmetry averaging explains the invariant component; Shannon entropy accounts for the distinctions that component does not display.

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